Decision Tree

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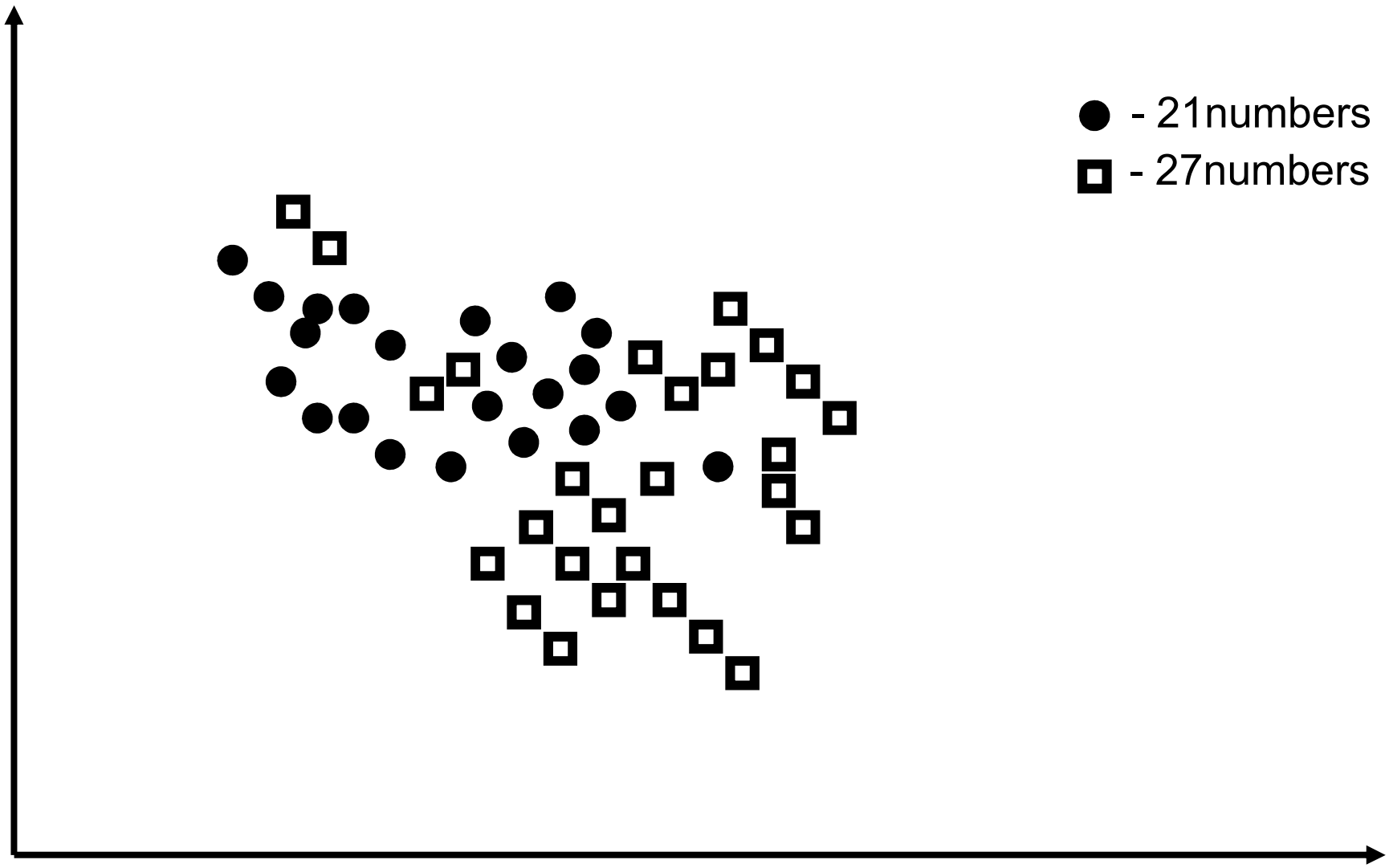
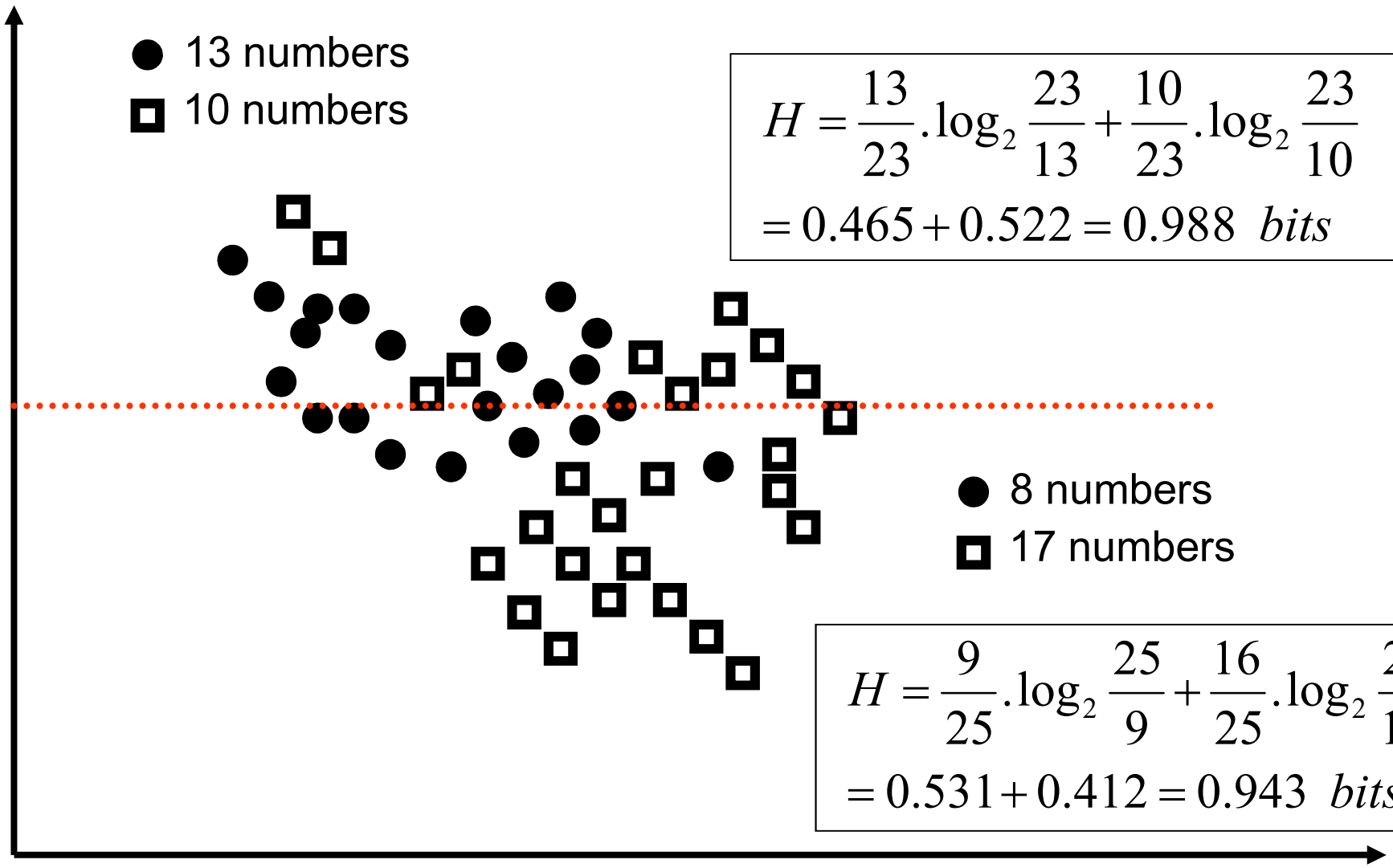
What is entropy?

- Measure of disorder
- Use it to classify the objects



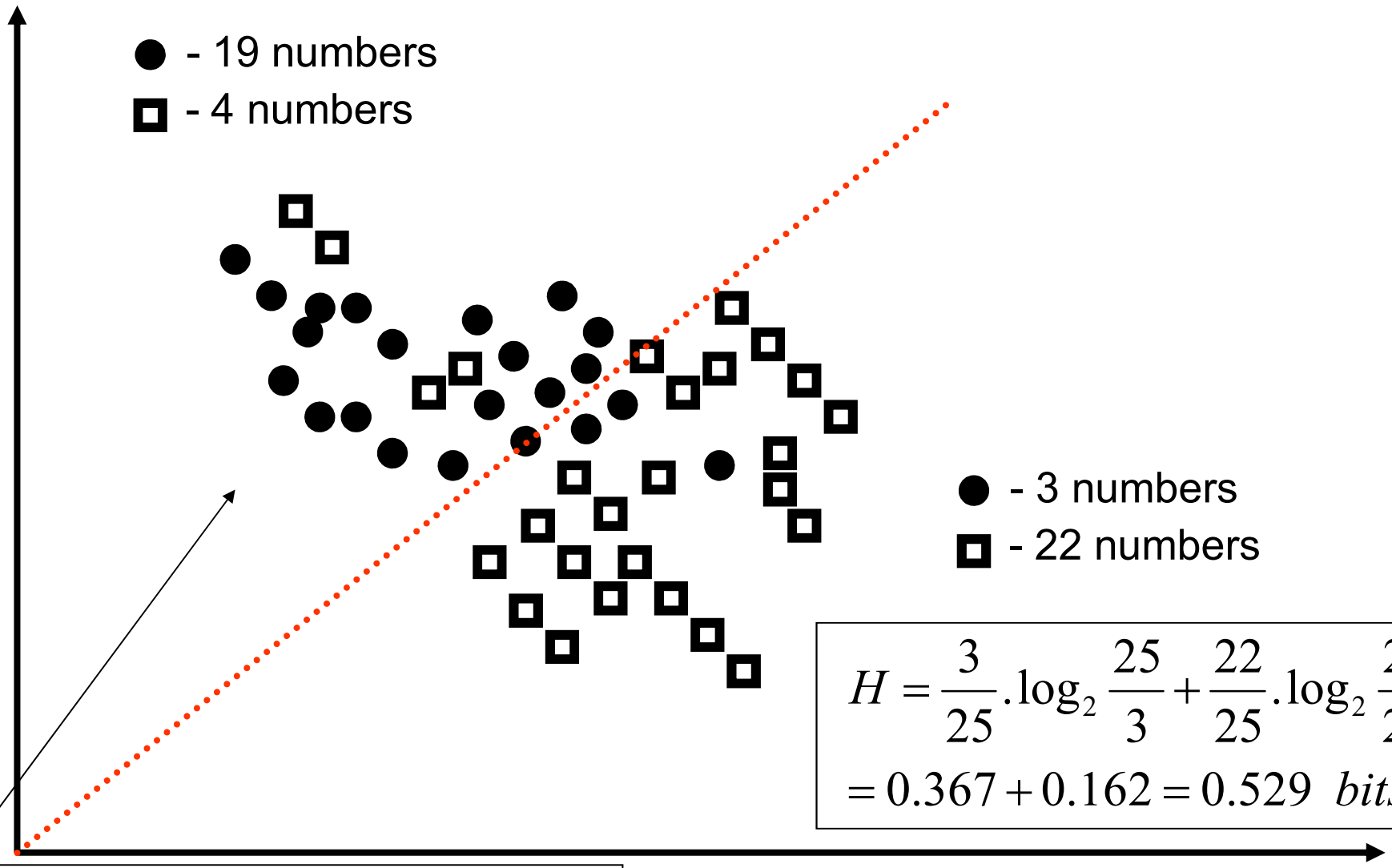
$$H = \frac{21}{48}.\log_2 \frac{48}{21} + \frac{27}{48}.\log_2 \frac{48}{27}$$

$$= 0.522 + 0.467 = 0.989 \ \ bits$$



$$H = \frac{13}{23}.\log_2 \frac{23}{13} + \frac{10}{23}.\log_2 \frac{23}{10}$$
$$= 0.465 + 0.522 = 0.988 \ \ bits$$

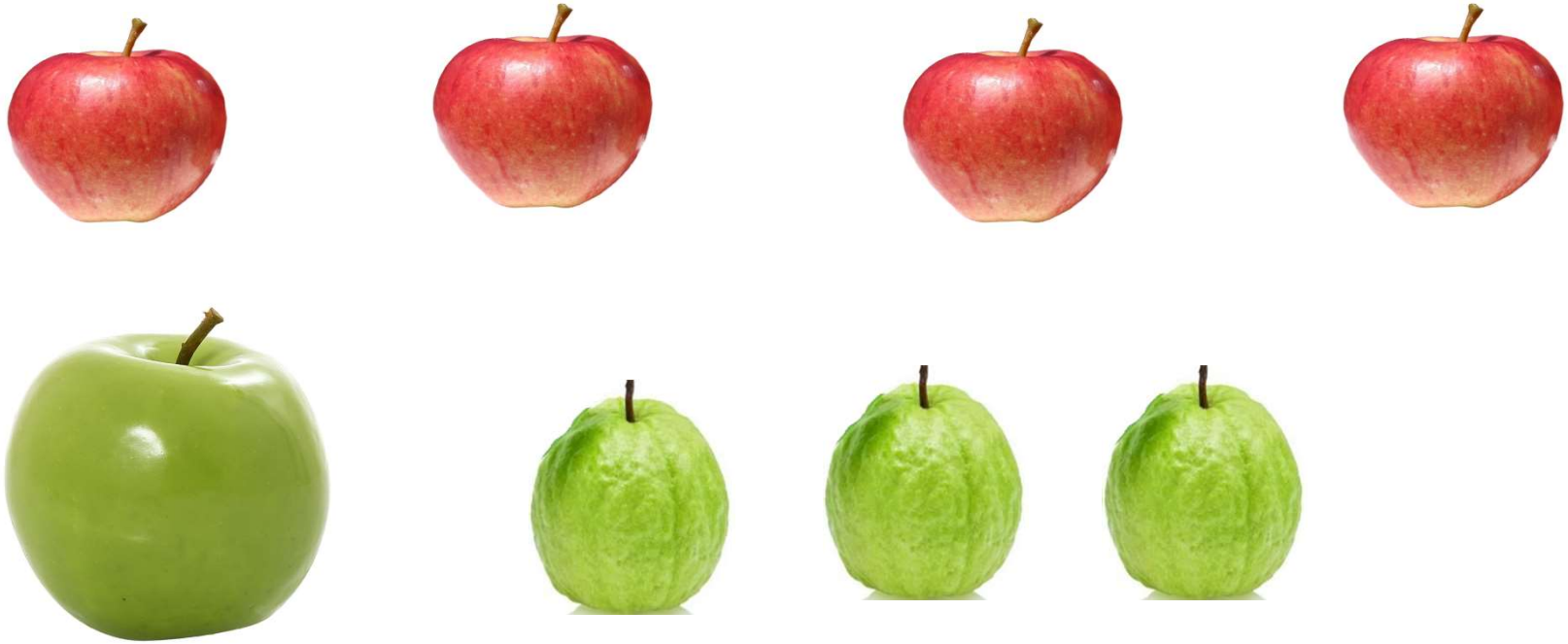
$$H = \frac{9}{25}.\log_2 \frac{25}{9} + \frac{16}{25}.\log_2 \frac{25}{16}$$
$$= 0.531 + 0.412 = 0.943 \ \ bits$$



$$H = \frac{3}{25}.\log_2 \frac{25}{3} + \frac{22}{25}.\log_2 \frac{25}{22}$$
$$= 0.367 + 0.162 = 0.529 \ \ bits$$

$$H = \frac{19}{23}.\log_2 \frac{23}{19} + \frac{4}{23}.\log_2 \frac{23}{4}$$
$$= 0.228 + 0.439 = 0.667 \ \ bits$$

What is entropy?



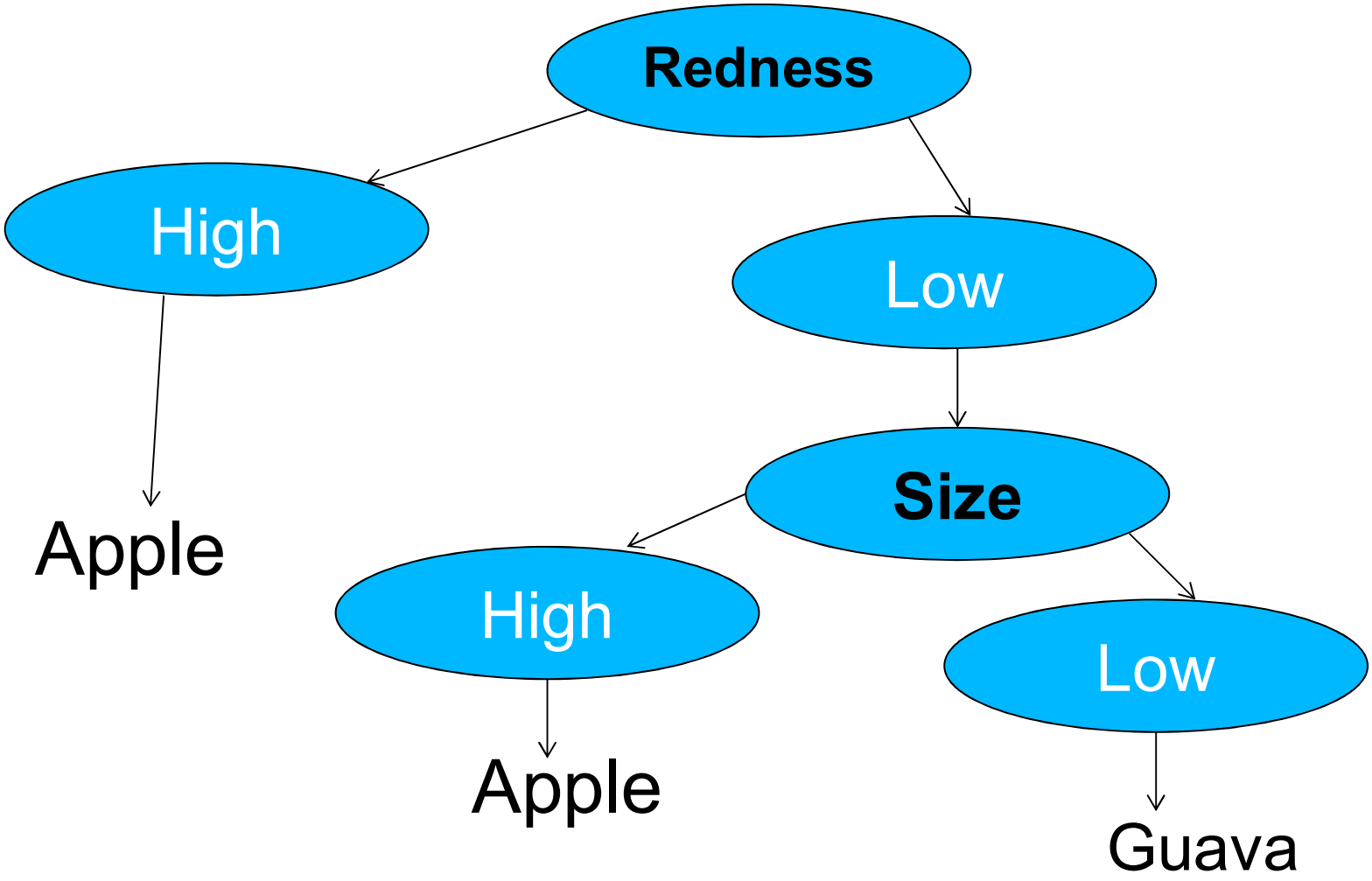
Observations

(1) Green apples are bigger in size compared to red apples

(2) Guavas are smaller than green apples but are equal to red apples

Redness	Size	Fruit	
High	Low	Apple	
Low	High	Apple	
Low	Low	Guava	

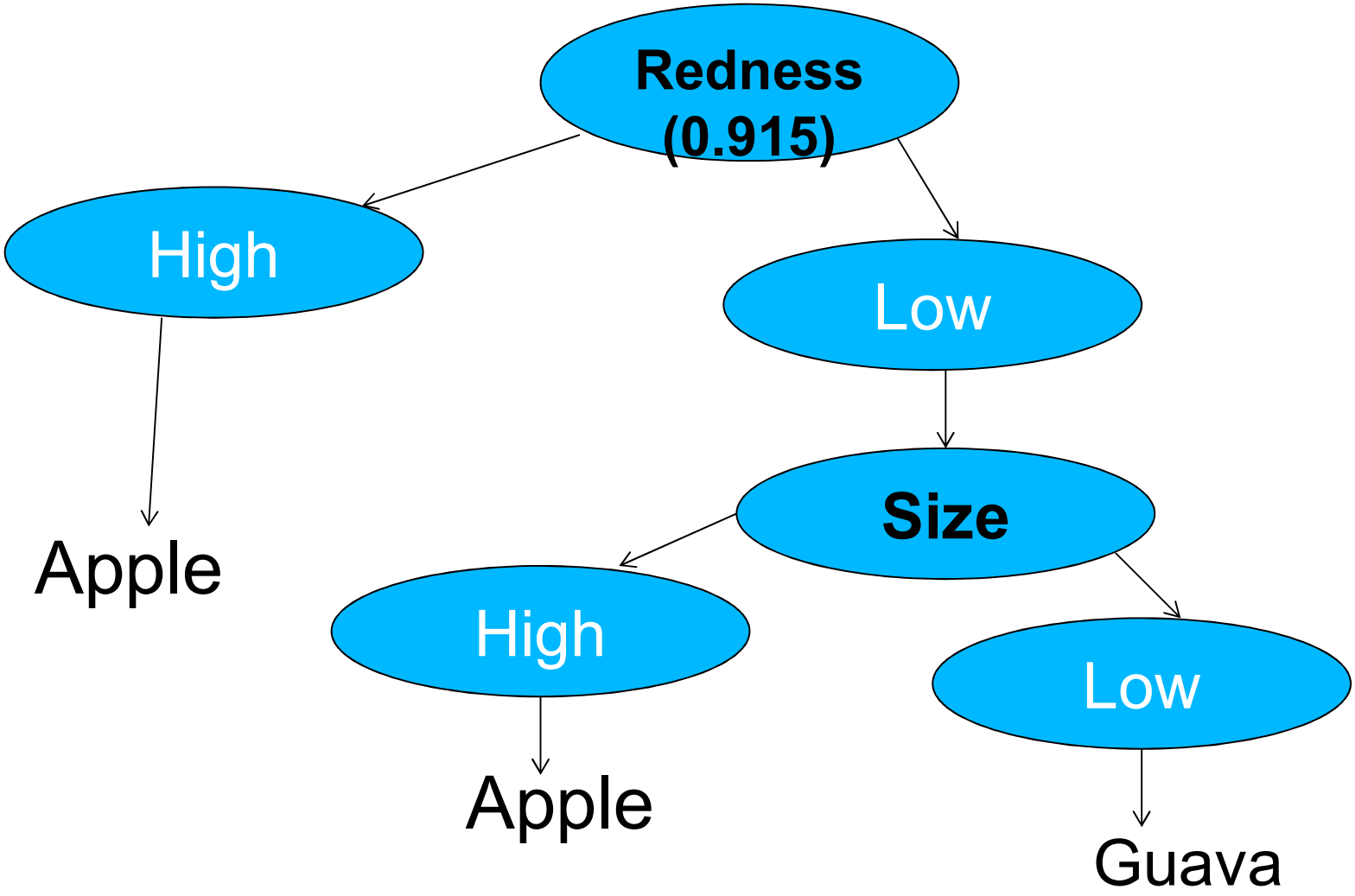
Building decision tree



Entropy of redness

- Two outcomes
  - Low redness
  - High redness
- P(low redness) = 2/3=0.67
- P(high redness) = 1/3=0.33
- H=-0.67*log(0.67) – 0.33*log(0.33) =0.915 bits

Building decision tree



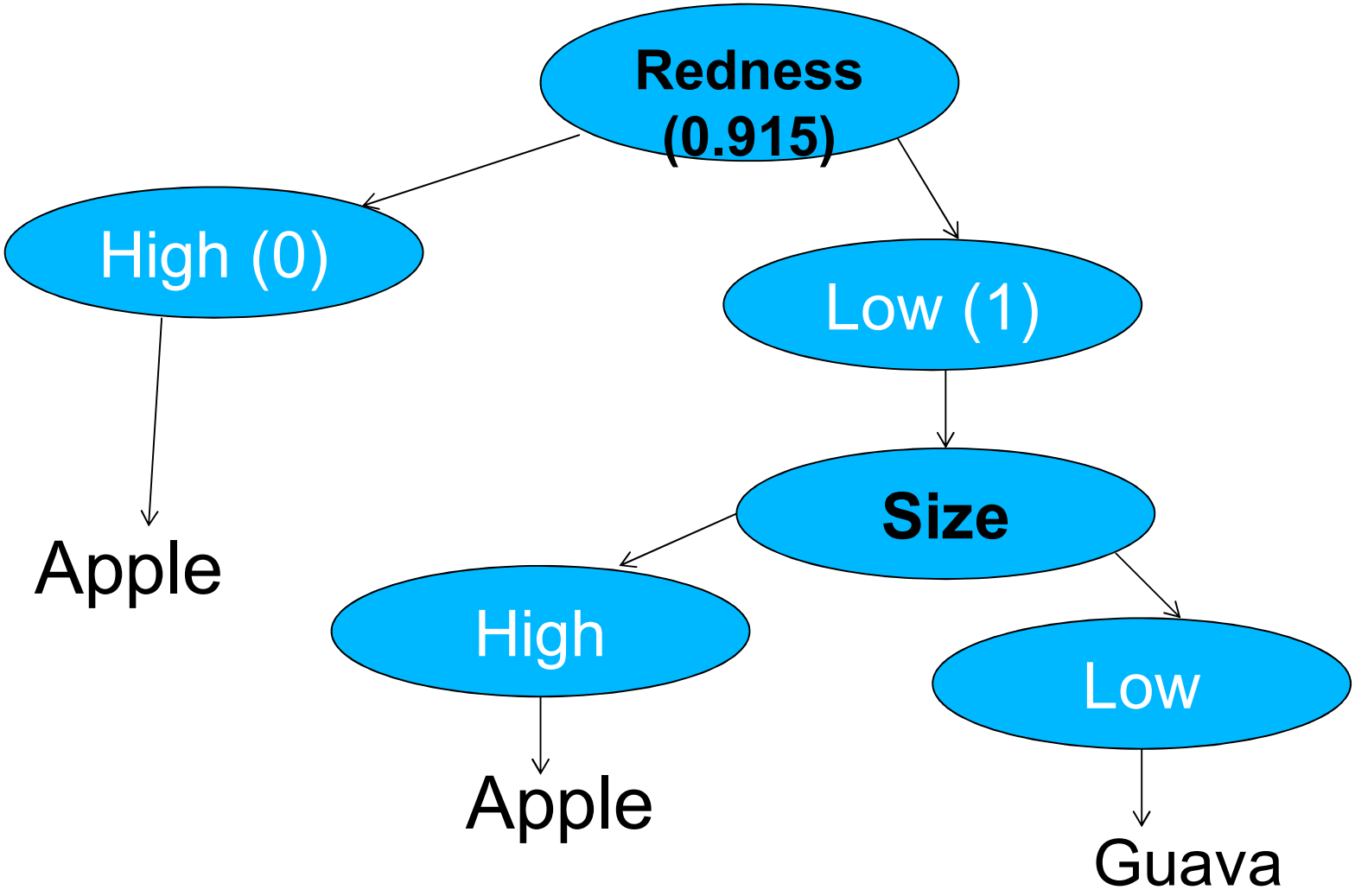
Entropy of High redness

- Only one object i.e. apple
- H = 0

Entropy of Low redness

- Could be either apple or guava
- H = -0.5*log(0.5) – 0.5*log(0.5) = 1 bit

Building decision tree

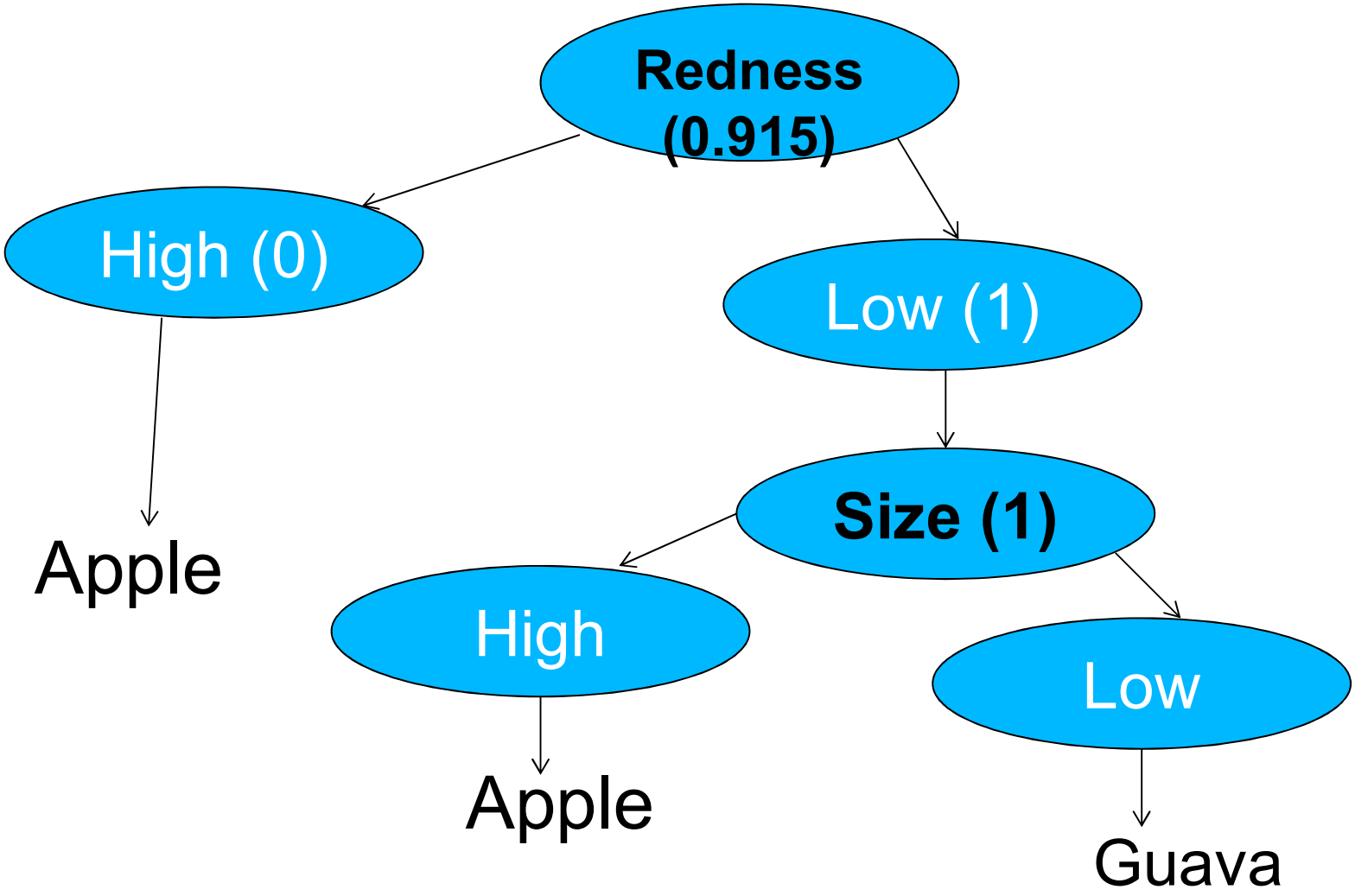


Entropy of size with low redness

- Two outcomes
  - Low size
  - High size
- P(low size) = 1/2=0.5
- P(high size) = 1/2=0.5
- H=-0.5*log(0.5) – 0.5*log(0.5)

  =1 bit

Building decision tree



Entropy of High size (with low redness)

- Only one object i.e. apple
- H = 0

Entropy of Low size (with low redness)

- Only one object i.e. Guava
- H = 0

One more example – make use of information gain

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Find H of E

Total = 16

8 positives

8 negatives

P(+ve) = P(-ve)=0.5

H(E) = -0.5*log(0.5)

-0.5*log(0.5) = 1 bit

Starting
(+ as well as)
H = 1

Based on A, B, C, D – we classify

- Randomly choose some number for A say 5
- No. of A ≥ 5= 12
- No. of A < 5= 4

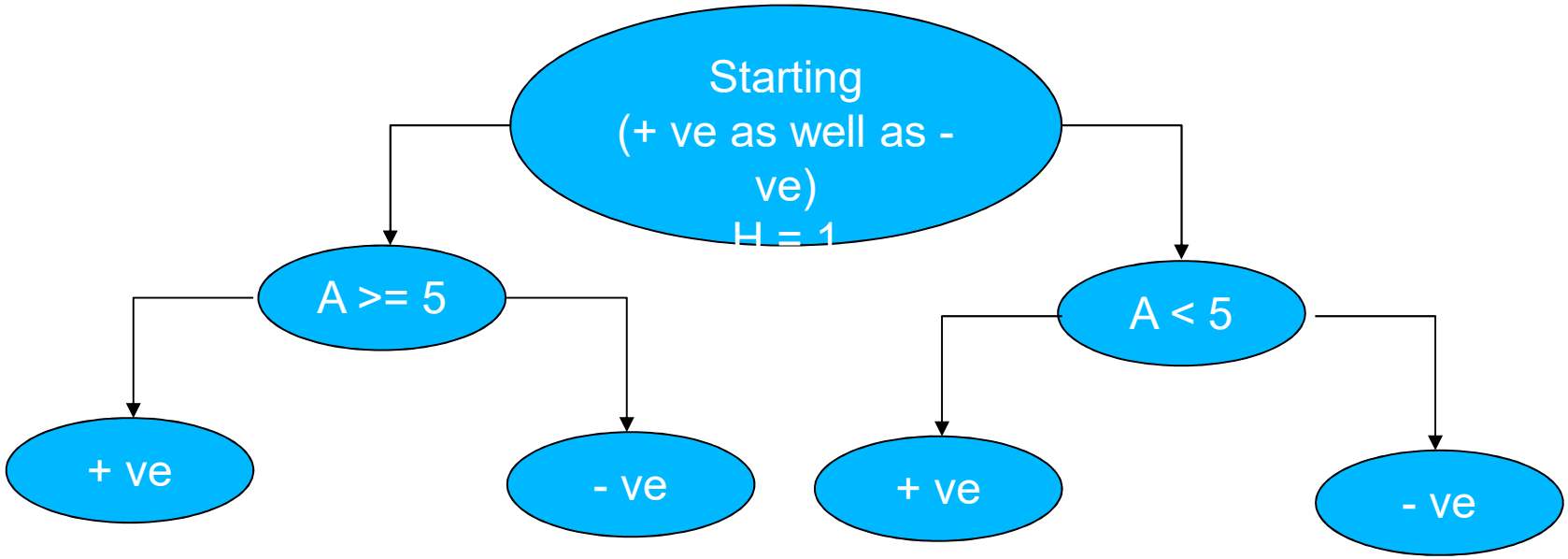
	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($A \geq 5$)

- No. of $A \geq 5 = 12$
- Some of them +ve and some of them –ve
- 5 positive
- 7 negative
- P(+ve with $A \geq 5$) = 5/12
- P(-ve with $A \geq 5$) = 7/12
- $H(A \geq 5) = 0.98$ bits

Entropy of ($A < 5$)

- No. of $A < 5 = 4$
- Some of them +ve and some of them –ve
- 3 positive
- 1 negative
- P(+ve with $A < 5$) = 3/4
- P(-ve with $A < 5$) = 1/4
- $H(A<5) = 0.81$ bits


Starting
(+ ve as well as -
ve)
H = 1
A >= 5
A < 5
+ ve
- ve
+ ve
- ve

Entropy of A

- Average of $H(A \geq 5)$ and $H(A < 5)$
- $P(A \geq 5)*H(A \geq 5) + P(A<5)*H(A<5)$
- $=12/16*0.98+4/16*0.81 = 0.938$ bits

- Information Gain$=H(E) - H(A)$

$=1-0.938=0.062$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for B say 3
- No. of B ≥ 3= 12
- No. of B < 3= 4

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($B \geq 3$)

- No. of $B \geq 3 = 12$
- Some of them +ve and some of them –ve
- 8 positive
- 4 negative
- P(+ve with $B \geq 3$) = 8/12
- P(-ve with $B \geq 3$) = 4/12
- $H(B \geq 3) = 0.915$ bits

Entropy of ($B < 3$)

- No. of $B < 3 = 4$
- Some of them +ve and some of them –ve
- 0 positive
- 4 negative
- P(+ve with $B < 3$) = 0
- P(-ve with $B < 3$) = 1
- $H(B<3)=0$

Entropy of B

- Average of $H(B \geq 3)$ and $H(B < 3)$
- $P(B \geq 3)*H(B \geq 3) + P(B<3)*H(B<3)$
- $=12/16*0.915+4/16*0 = 0.687$ bits

- Information Gain$=H(E) - H(B)$

$=1-0.687=0.313$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for C say 4.2
- No. of C ≥ 4.2= 6
- No. of C< 4.2 = 10

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($C \geq 5$)

- No. of $C \geq 4.2 = 6$
- Some of them +ve and some of them –ve
- 0 positive
- 6 negative
- P(+ve with $C \geq 4.2$) = 0/6
- P(-ve with $C \geq 4.2$) = 6/6
- $H(C \geq 4.2)=0$

Entropy of ($C < 5$)

- No. of $C < 4.2 = 10$
- Some of them +ve and some of them –ve
- 8 positive
- 2 negative
- P(+ve with $C < 4.2$) = 8/10
- P(-ve with $C < 4.2$) = 2/10
- $H(C < 4.2)=0.723$ bits

Entropy of C

- Average of $H(C \geq 4.2)$ and $H(C < 4.2)$
- $P(C \geq 4.2)*H(C \geq 4.2) + P(C< 4.2)*H(C< 4.2)$
- $=6/16*0+10/16*0.723 = 0.452$ bits

- Information Gain$=H(E) - H(C)$

$=1-0.452=0.548$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for D say 1.4
- No. of D ≥ 1.4= 5
- No. of D< 1.4 = 11

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($D \geq 1.4$)

- No. of $D \geq 1.4 = 5$
- Some of them +ve and some of them –ve
- 0 positive
- 5 negative
- P(+ve with $D \geq 1.4$) = 0/5
- P(-ve with $D \geq 1.4$) = 5/5
- $H(D \geq 1.4)=0$

Entropy of ($D < 1.4$)

- No. of $D < 1.4 = 11$
- Some of them +ve and some of them –ve
- 8 positive
- 3 negative
- P(+ve with $D<1.4$) = 8/11
- P(-ve with $D<1.4$) = 3/11
- $H(D < 1.4)=0.846$ bits

Entropy of D

- Average of $H(D \geq 1.4)$ and $H(D < 1.4)$
- $P(D \geq 4.2)*H(D \geq 1.4) + P(D<1.4)*H(D<1.4)$
- $=5/16*0+11/16*0.846 = 0.582$ bits

- Information Gain$=H(E) - H(D)$

$=1-0.582=0.418$ bits

Order A, B, C, D based on information gain

IG of A = 0.062 bits

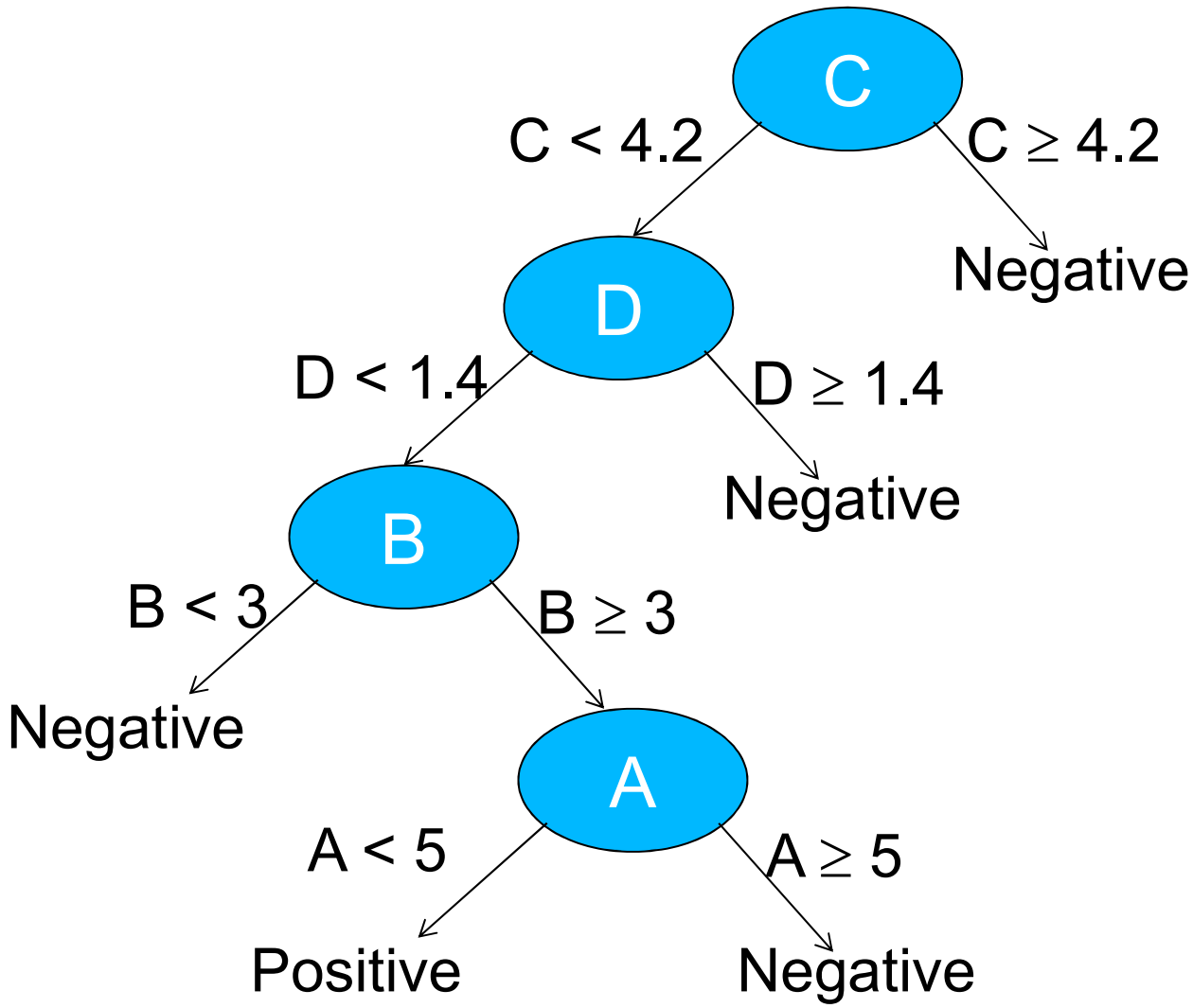
IG of B = 0.313 bits

IG of C = 0.548 bits

IG of D = 0.418 bits

Rank them

C, D, B, A


C
C < 4.2
C ≥ 4.2
Negative
D
D < 1.4
D ≥ 1.4
Negative
B
B < 3
B ≥ 3
Negative
A
A < 5
A ≥ 5
Positive
Negative

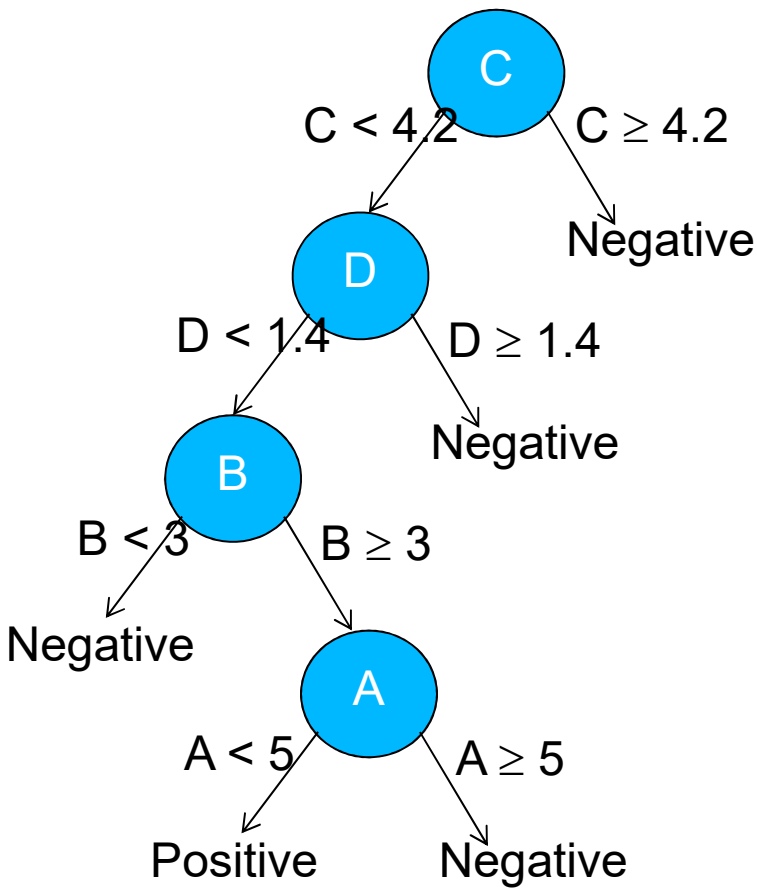
Decision Tree algorithm

- Target
- Target entropy
- Variables
- Variable entropy
- Information gain
- Ranking
- Drawing tree

Rule based classification

What is rule?

If ***something*** Then ***something***



Decision Tree ≡ Rule-based

If C >=4.2

then conclude negative

If C < 4.2 & D >=1.4

then conclude negative

If C < 4.2 & D < 1.4 & B >=3

then conclude negative

If C < 4.2 & D < 1.4 & B < 3

then conclude negative

...

...

Not Unique

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Find H of E

Total = 16

8 positives

8 negatives

P(+ve) = P(-ve)=0.5

H(E) = -0.5*log(0.5)

-0.5*log(0.5) = 1 bit

Based on A, B, C, D – we classify

- Randomly choose some number for A say 6.5
- No. of A ≥ 6.5= 3
- No. of A < 6.5= 13

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($A \geq 6.5$)

- No. of $A \geq 6.5 = 3$
- Some of them +ve and some of them –ve
- 0 positive
- 3 negative
- P(+ve with $A \geq 6.5$) = 0
- P(-ve with $A \geq 6.5$) = 1
- $H(A \geq 6.5)=0$

Entropy of ($A < 6.5$)

- No. of $A < 6.5 = 13$
- Some of them +ve and some of them –ve
- 8 positive
- 5 negative
- P(+ve with $A < 6.5$) = 8/13
- P(-ve with $A < 6.5$) = 5/13
- $H(A<6.5)=0.96$ bits

Entropy of A

- Average of $H(A \geq 6.5)$ and $H(A < 6.5)$
- $P(A \geq 6.5)*H(A \geq 6.5) + P(A<6.5)*H(A<6.5)$
- $=3/16*0+13/16*0.96 = 0.78$ bits

- Information Gain$=H(E) - H(A)$

$=1-0.78=0.22$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for B say 3
- No. of B ≥ 3= 12
- No. of B < 3= 4

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($B \geq 3$)

- No. of $B \geq 3 = 12$
- Some of them +ve and some of them –ve
- 8 positive
- 4 negative
- P(+ve with $B \geq 3$) = 8/12
- P(-ve with $B \geq 3$) = 4/12
- $H(B \geq 3) = 0.915$ bits

Entropy of ($B < 3$)

- No. of $B < 3 = 4$
- Some of them +ve and some of them –ve
- 0 positive
- 4 negative
- P(+ve with $B < 3$) = 0
- P(-ve with $B < 3$) = 1
- $H(B<3)=0$

Entropy of B

- Average of $H(B \geq 3)$ and $H(B < 3)$
- $P(B \geq 3)*H(B \geq 3) + P(B<3)*H(B<3)$
- $=12/16*0.915+4/16*0 = 0.687$ bits

- Information Gain$=H(E) - H(B)$

$=1-0.687=0.313$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for C say 4.2
- No. of C ≥ 4.2= 6
- No. of C< 4.2 = 10

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($C \geq 4.2$)

- No. of $C \geq 4.2 = 6$
- Some of them +ve and some of them –ve
- 0 positive
- 6 negative
- P(+ve with $C \geq 4.2$) = 0/6
- P(-ve with $C \geq 4.2$) = 6/6
- $H(C \geq 4.2)=0$

Entropy of ($C < 4.2$)

- No. of $C < 4.2 = 10$
- Some of them +ve and some of them –ve
- 8 positive
- 2 negative
- P(+ve with $C < 4.2$) = 8/10
- P(-ve with $C < 4.2$) = 2/10
- $H(C < 4.2)=0.723$ bits

Entropy of C

- Average of $H(C \geq 4.2)$ and $H(C < 4.2)$
- $P(C \geq 4.2)*H(C \geq 4.2) + P(C< 4.2)*H(C< 4.2)$
- $=6/16*0+10/16*0.723 = 0.452$ bits

- Information Gain$=H(E) - H(C)$

$=1-0.452=0.548$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for D say 1.4
- No. of D ≥ 1.4= 5
- No. of D< 1.4 = 11

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($D \geq 1.4$)

- No. of $D \geq 1.4 = 5$
- Some of them +ve and some of them –ve
- 0 positive
- 5 negative
- P(+ve with $D \geq 1.4$) = 0/5
- P(-ve with $D \geq 1.4$) = 5/5
- $H(D \geq 1.4)=0$

Entropy of ($D < 1.4$)

- No. of $D < 1.4 = 11$
- Some of them +ve and some of them –ve
- 8 positive
- 3 negative
- P(+ve with $D<1.4$) = 8/11
- P(-ve with $D<1.4$) = 3/11
- $H(D < 1.4)=0.846$ bits

Entropy of D

- Average of $H(D \geq 1.4)$ and $H(D < 1.4)$
- $P(D \geq 4.2)*H(D \geq 1.4) + P(D<1.4)*H(D<1.4)$
- $=5/16*0+11/16*0.846 = 0.582$ bits

- Information Gain$=H(E) - H(D)$

$=1-0.582=0.418$ bits

Order A, B, C, D based on information gain

IG of A = 0.22 bits

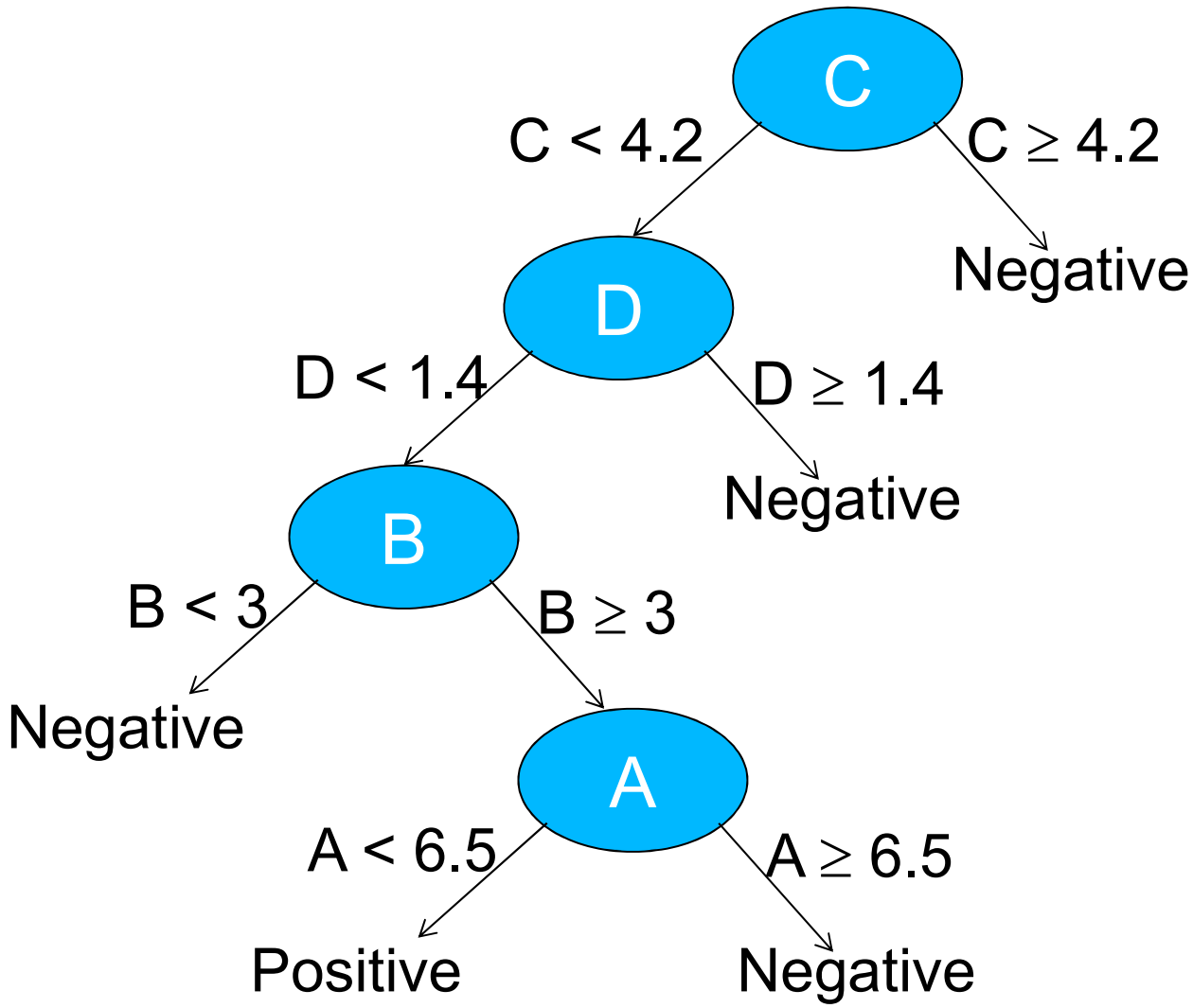
IG of B = 0.313 bits

IG of C = 0.548 bits

IG of D = 0.418 bits

Rank them

C, D, B, A


C
C < 4.2
C ≥ 4.2
Negative
D
D < 1.4
D ≥ 1.4
Negative
B
B < 3
B ≥ 3
Negative
A
A < 6.5
A ≥ 6.5
Positive
Negative

Not Unique

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Find H of E

Total = 16

8 positives

8 negatives

P(+ve) = P(-ve)=0.5

H(E) = -0.5*log(0.5)

-0.5*log(0.5) = 1 bit

Based on A, B, C, D – we classify

- Randomly choose some number for A say 6.5
- No. of A ≥ 6.5= 3
- No. of A < 6.5= 13

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($A \geq 6.5$)

- No. of $A \geq 6.5 = 3$
- Some of them +ve and some of them –ve
- 0 positive
- 3 negative
- P(+ve with $A \geq 6.5$) = 0
- P(-ve with $A \geq 6.5$) = 1
- $H(A \geq 6.5)=0$

Entropy of ($A < 6.5$)

- No. of $A < 6.5 = 13$
- Some of them +ve and some of them –ve
- 8 positive
- 5 negative
- P(+ve with $A < 6.5$) = 8/13
- P(-ve with $A < 6.5$) = 5/13
- $H(A<6.5)=0.96$ bits

Entropy of A

- Average of $H(A \geq 6.5)$ and $H(A < 6.5)$
- $P(A \geq 6.5)*H(A \geq 6.5) + P(A<6.5)*H(A<6.5)$
- $=3/16*0+13/16*0.96 = 0.78$ bits

- Information Gain$=H(E) - H(A)$

$=1-0.78=0.22$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for B say 2.4
- No. of B ≥ 2.4= 15
- No. of B < 2.4= 1

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($B \geq 2.4$)

- No. of $B \geq 2.4 = 15$
- Some of them +ve and some of them –ve
- 8 positive
- 7 negative
- P(+ve with $B \geq 2.4$) = 8/15
- P(-ve with $B \geq 2.4$) = 7/15
- $H(B \geq 2.4) = 0.997$ bits

Entropy of ($B < 2.4$)

- No. of $B < 2.4 = 1$
- Some of them +ve and some of them –ve
- 0 positive
- 1 negative
- P(+ve with $B < 2.4$) = 0
- P(-ve with $B < 2.4$) = 1
- $H(B<2.4)=0$

Entropy of B

- Average of $H(B \geq 2.4)$ and $H(B < 2.4)$
- $P(B \geq 2.4)*H(B \geq 2.4) + P(B<2.4)*H(B<2.4)$
- $=15/16*0.915+1/16*0 = 0.857$ bits

- Information Gain$=H(E) - H(B)$

$=1-0.857=0.143$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for C say 4.2
- No. of C ≥ 4.2= 6
- No. of C< 4.2 = 10

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($C \geq 4.2$)

- No. of $C \geq 4.2 = 6$
- Some of them +ve and some of them –ve
- 0 positive
- 6 negative
- P(+ve with $C \geq 4.2$) = 0/6
- P(-ve with $C \geq 4.2$) = 6/6
- $H(C \geq 4.2)=0$

Entropy of ($C < 4.2$)

- No. of $C < 4.2 = 10$
- Some of them +ve and some of them –ve
- 8 positive
- 2 negative
- P(+ve with $C < 4.2$) = 8/10
- P(-ve with $C < 4.2$) = 2/10
- $H(C < 4.2)=0.723$ bits

Entropy of C

- Average of $H(C \geq 4.2)$ and $H(C < 4.2)$
- $P(C \geq 4.2)*H(C \geq 4.2) + P(C< 4.2)*H(C< 4.2)$
- $=6/16*0+10/16*0.723 = 0.452$ bits

- Information Gain$=H(E) - H(C)$

$=1-0.452=0.548$ bits

Based on A, B, C, D – we classify

- Randomly choose some number for D say 1.4
- No. of D ≥ 1.4= 5
- No. of D< 1.4 = 11

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	4.9	2.4	3.3	1	negative

Entropy of ($D \geq 1.4$)

- No. of $D \geq 1.4 = 5$
- Some of them +ve and some of them –ve
- 0 positive
- 5 negative
- P(+ve with $D \geq 1.4$) = 0/5
- P(-ve with $D \geq 1.4$) = 5/5
- $H(D \geq 1.4)=0$

Entropy of ($D < 1.4$)

- No. of $D < 1.4 = 11$
- Some of them +ve and some of them –ve
- 8 positive
- 3 negative
- P(+ve with $D<1.4$) = 8/11
- P(-ve with $D<1.4$) = 3/11
- $H(D < 1.4)=0.846$ bits

Entropy of D

- Average of $H(D \geq 1.4)$ and $H(D < 1.4)$
- $P(D \geq 4.2)*H(D \geq 1.4) + P(D<1.4)*H(D<1.4)$
- $=5/16*0+11/16*0.846 = 0.582$ bits

- Information Gain$=H(E) - H(D)$

$=1-0.582=0.418$ bits

Order A, B, C, D based on information gain

IG of A = 0.22 bits

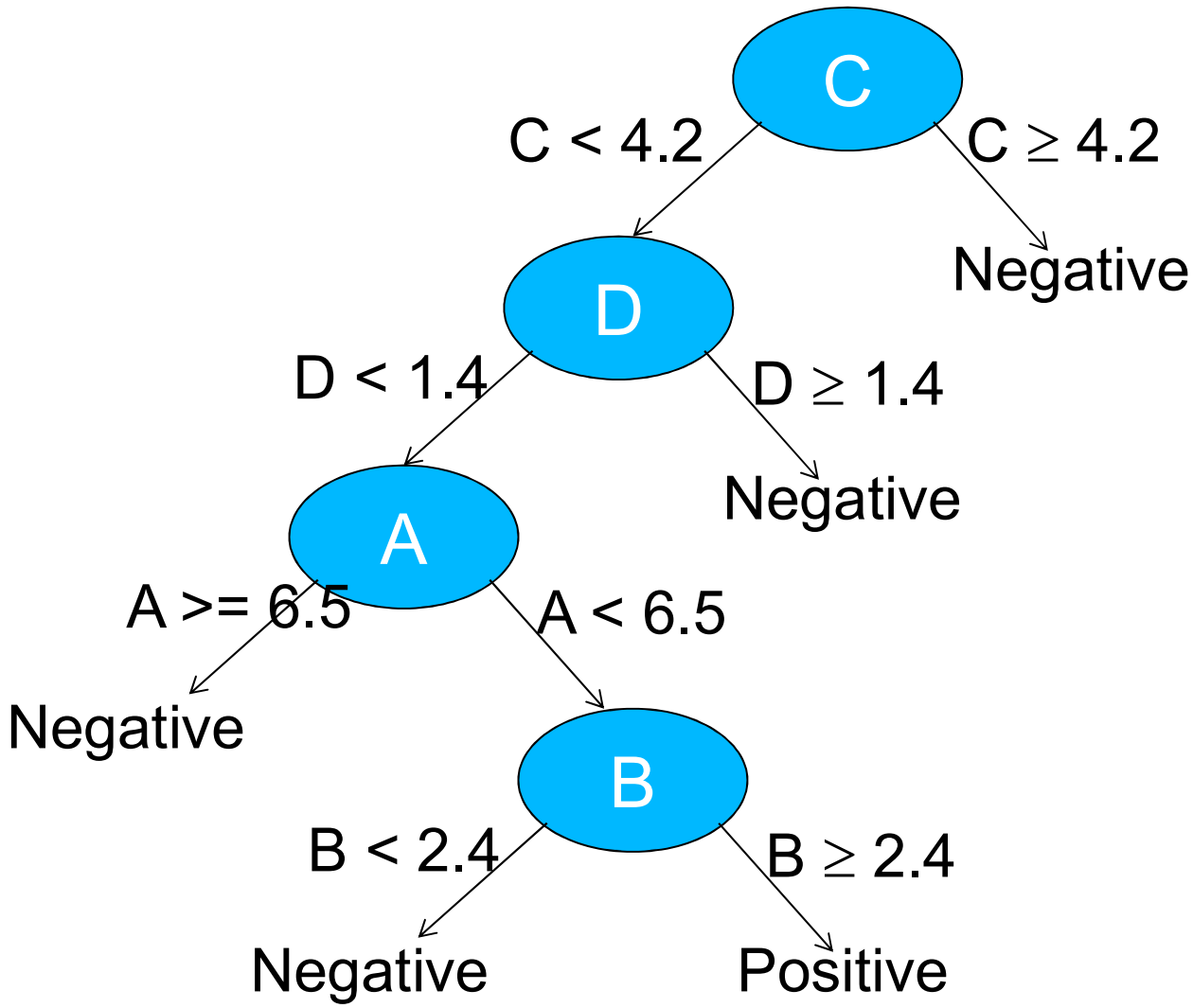
IG of B = 0.143 bits

IG of C = 0.548 bits

IG of D = 0.418 bits

Rank them

C, D, A, B


C
C < 4.2
C ≥ 4.2
Negative
D
D < 1.4
D ≥ 1.4
Negative
A
A >= 6.5
A < 6.5
Negative
B
B < 2.4
B ≥ 2.4
Negative
Positive

What'll happen here?

	A	B	C	D	E
1	4.8	3.4	1.9	0.2	positive
2	5	3	1.6	0.2	positive
3	5	3.4	1.6	0.4	positive
4	5.2	3.5	1.5	0.2	positive
5	5.2	3.4	1.4	0.2	positive
6	4.7	3.2	1.6	0.2	positive
7	4.8	3.1	1.6	0.2	positive
8	5.4	3.4	1.5	0.4	positive
9	7	3.2	4.7	1.4	negative
10	6.4	3.2	4.5	1.5	negative
11	6.9	3.1	4.9	1.5	negative
12	5.5	2.3	4	1.3	negative
13	6.5	2.8	4.6	1.5	negative
14	5.7	2.8	4.5	1.3	negative
15	6.3	3.3	4.7	1.6	negative
16	6.3	3.3	1.5	1.6	negative

One more example

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Entropy of "Play Golf"

Target

Play Golf
No
No
Yes
Yes
Yes
No
Yes
No
Yes
Yes
Yes
Yes
Yes
No

$$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5

Entropy(PlayGolf) = Entropy (5,9)

= Entropy (0.36, 0.64)

= - (0.36 $\log_2$ 0.36) - (0.64 $\log_2$ 0.64)

= 0.94

$$E(T,X) = \sum_{c \in X} P(c)E(c)$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

E(PlayGolf, Outlook) = **P**(Sunny)***E**(3,2) + **P**(Overcast)***E**(4,0) + **P**(Rainy)***E**(2,3)

= (5/14)*0.971 + (4/14)*0.0 + (5/14)*0.971

= 0.693

Information gain

Entropy(Play Golf) – Entropy(Play Golf, Outlook)

0.94 – 0.693 = 0.247

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1
Gain = 0.029			

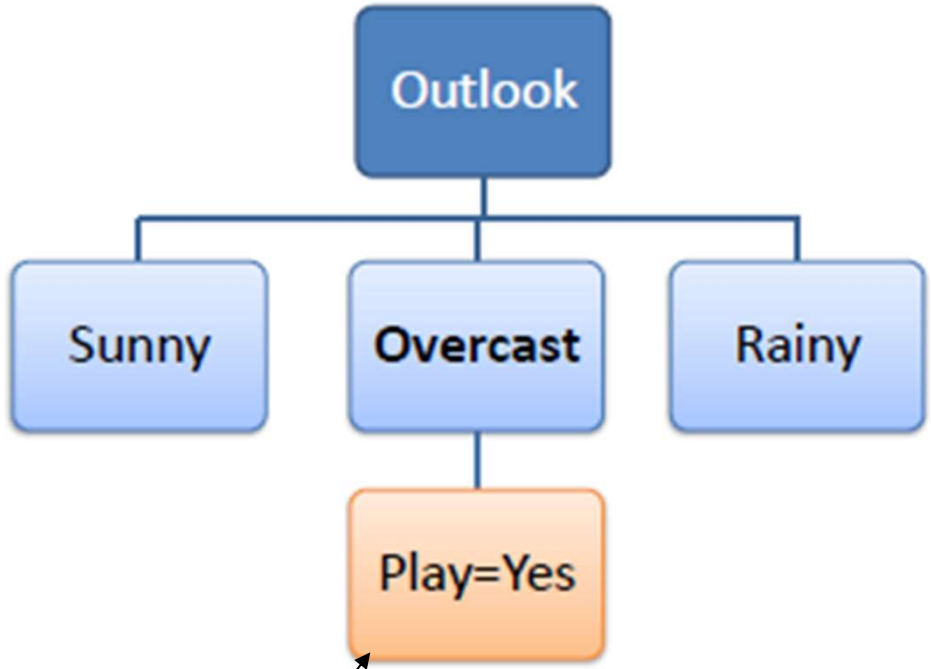
		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1
Gain = 0.152			

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3
Gain = 0.048			

Go for highest gain i.e. Outlook

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			



Leaf node (because, entropy =0)
(no further action needed)

Under sunny we have 3 more parameters

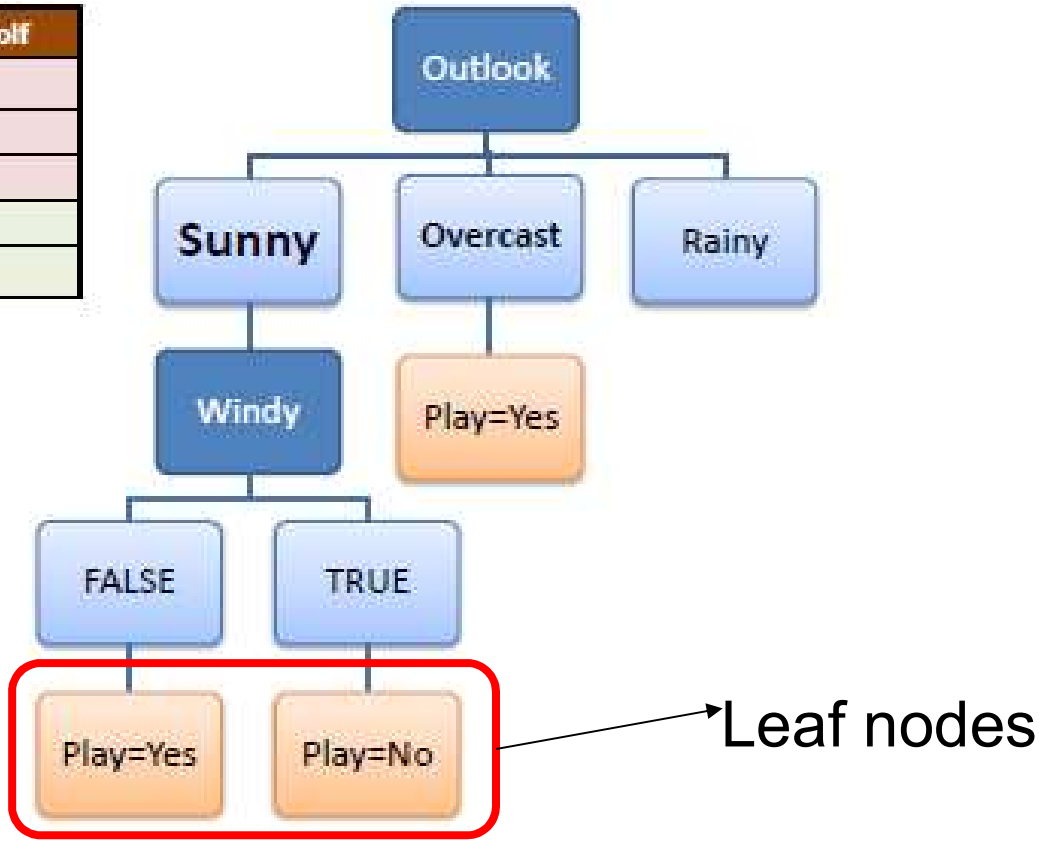
Temp	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	Normal	FALSE	Yes
Mild	High	TRUE	No

Find gain & go for maximum
In this case, windy.

Entropy(3/5,2/5)

Entropy(3/5,2/5)

Temp	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Mild	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	High	TRUE	No



		Play Golf	
		Yes	No
Temp	Mild	2	1
	Cool	1	1
Gain=			

		Play Golf	
		Yes	No
Humidity	Normal	2	1
	High	1	1
Gain =			

		Play Golf	
		Yes	No
Windy	False	3	0
	True	0	2
Gain =			

Hot	High	FALSE	No
Hot	High	TRUE	No
Mild	High	FALSE	No
Cool	Normal	FALSE	Yes
Mild	Normal	TRUE	Yes

		Play Golf	
		Yes	No
Temp	Mild	1	1
	Hot	0	2
	Cool	1	0
Gain=			

		Play Golf	
		Yes	No
Humidity	Normal	2	0
	High	0	3
Gain =			

		Play Golf	
		Yes	No
Windy	False	1	2
	True	1	1
Gain =			